

Machine Learning-Assisted Crystal Structure Prediction of Solid-State Electrolytes Reveals Superior Ionic Conductivity in Metastable Edge-Sharing Phases

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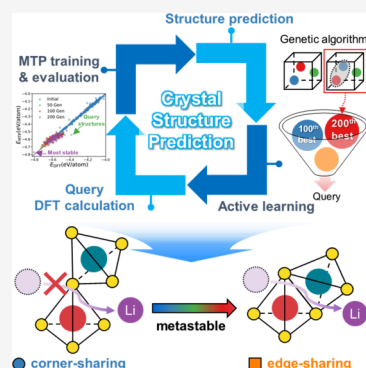


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ABSTRACT: Significant attention has been devoted to developing novel solid-state electrolytes (SSEs) with high ionic conductivity for all-solid-state batteries (ASSBs). However, most studies have primarily focused on compositional substitutions, often overlooking the fundamental role of inherent crystal structures on ion transport. To address this, we introduce a theoretical crystal structure prediction (CSP) approach based on the machine-learning moment tensor potential (MTP). The proposed approach successfully identifies novel SSE structures and reproduces 12 experimental crystal structures. Using a phase-diagram-guided strategy, CSP is applied to four promising SSE candidates, Li_2SiS_3 , Li_2GeS_3 , $\text{Li}_4\text{SiGeS}_6$, and $\text{Li}_4\text{SiSnS}_6$, to assess their polyhedral connectivity, relative stability, and Li-ion transport properties. The results reveal that metastable edge-sharing phases exhibit superior Li-ion mobility compared with their stable corner-sharing counterparts. This superior conductivity is attributed to the Li-ion accessible volume, quantified by the packing ratio (fraction of the unit cell volume occupied by nonconductive volume) and by the dynamic distortion of the Li-S_4 sublattice, which represents the local environment encountered by migrating Li-ions. The metastable phases feature higher packing efficiency, larger Li-S_4 sublattice volume, and greater distortion, all of which contribute to improved Li-ion transport. This study highlights the potential of CSP to design novel SSEs and high-performance ASSBs.



1. INTRODUCTION

To pursue safer and more energy-dense lithium-ion batteries (LIBs), all-solid-state batteries (ASSBs) have garnered significant attention as next-generation batteries.^{1–3} Specifically, ASSBs replace the flammable and volatile liquid-electrolytes used in conventional LIBs with solid-state electrolytes (SSEs),^{4–6} thereby addressing safety concerns such as the solvent's flammability and volatilization.⁷ Furthermore, the potential use of Li metal, instead of conventional graphite anodes offers a significant increase in energy density, enhancing the overall efficiency of ASSBs.^{4,5} Notably, SSEs must satisfy multiple requirements, including high ionic conductivity, electrochemical stability, and interfacial compatibility. To meet these demands, extensive research has focused on the development of multicomponent inorganic SSEs through compositional optimization, typically via elemental substitution or doping with base materials such as $\text{Li}_6\text{PS}_5\text{Cl}$ (argyrodite),^{8–11} $\text{Li}_7\text{P}_3\text{S}_{11}$,^{12,13} Li_3PS_4 ,^{14,15} and $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ (LGPS).¹⁶ Although this strategy has led to incremental improvements, it often fails to induce meaningful modifications in the underlying crystal structure, which is a key factor governing ion conductivity and overall material performance. The potential impact of such structural variations on SSE properties during such optimization processes has not been systematically investigated. Therefore, to enable substantial advances in SSE design, a more fundamental strategy is

required, namely, the systematic exploration and design of novel crystal structures that inherently promote superior ion transport characteristics.

With advancements in experimental synthesis techniques and instrumentation, a wide range of crystal structures have been accumulated. The Inorganic Crystal Structure Database (ICSD)^{17,18} contains over 200,000 experimentally validated structures, while theoretical databases, such as the Materials Project (MP),¹⁹ the Open Quantum Materials Database (OQMD),²⁰ and Automatic-Flow (AFLOW),²¹ provide numerous structures based on first-principles calculations. Despite these developments, many potentially important crystal structures remain undiscovered, as experimental discovery often relies on inefficient trial-and-error methods. In contrast, theoretical crystal structure prediction (CSP) offers a promising pathway for the targeted discovery of novel crystal structures based on chemical composition. In this context, phase-diagram-guided CSP approaches have emerged as effective tools for identifying promising compositional

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spaces wherein structurally distinct and metastable phases may arise. These approaches not only expand the accessible structural degrees of freedom through compositional tuning but also provide a predictive framework for discovering novel SSE phases with improved ion transport and electrochemical performance.^{22,23} On this basis, we applied the phase-diagram-guided CSP method to the Li_2SiS_3 compositional system, incorporating Group 14 element substitutions (Ge and Sn) to investigate four plausible candidates: Li_2SiS_3 , Li_2GeS_3 , $\text{Li}_4\text{SiGeS}_6$, and $\text{Li}_4\text{SiSnS}_6$. Subsequently, CSP was performed on these candidate systems, yielding a diverse set of metastable phases.

While theoretical CSP based on density functional theory (DFT) has been successfully applied to many material systems,^{24–31} using evolutionary algorithms such as particle swarm optimization,³² genetic algorithms,³³ and random structure search,³⁴ its broader applicability remains constrained by high computational cost, particularly for large or compositionally complex systems. To overcome these limitations, recent advances in machine-learned interatomic potentials (MLIPs) have enabled CSP with DFT-level accuracy but at a significantly lower computational cost. Consequently, more extensive evolutionary steps and larger compositional spaces can be investigated by integrating MLIPs with CSP frameworks. For example, Google recently applied this approach to screen over 2.2 million hypothetical structures, identifying hundreds of previously unreported but experimentally synthesized phases, including promising SSE candidates.³⁵ These developments have established MLIP-assisted CSP as a powerful tool that accelerates the discovery of new crystal structures and functional materials, especially within multi-component SSE systems.

To validate the proposed MLIP-assisted CSP framework, we applied it to 14 known experimental Li- or Na- containing crystal structures. The method successfully reproduced 12 of these structures, either by correctly identifying their atomic configurations or by predicting alternative structures with improved potential stability, demonstrating the framework's predictive reliability. Additionally, the proposed phase-diagram-guided and MLIP-assisted CSP successfully predicted structurally distinct metastable phases in the Li_2SiS_3 , Li_2GeS_3 , $\text{Li}_4\text{SiGeS}_6$, and $\text{Li}_4\text{SiSnS}_6$ systems, where 10 low-energy polymorphs exhibiting three representative $[\text{MS}_4]$ polyhedral topologies were identified: corner-sharing, edge-sharing, and mixed-type topologies ($M = \text{Si, Ge, Sn}$). Notably, the proposed CSP strategy successfully predicted the edge-sharing metastable phase of Li_2SiS_3 , previously identified by Kanno and co-workers under modified synthesis conditions and shown to exhibit ionic conductivity approximately 3 orders of magnitude higher than its conventional corner-sharing counterpart.³⁶ This consistency with experimental findings reinforces the credibility and practical relevance of our approach and highlights the effectiveness of the CSP strategy in uncovering novel solid-state electrolyte structures with enhanced functional potential.

To evaluate the impact of discovered structural motifs on Li-ion transport, we performed ab initio molecular dynamics (AIMD) simulations. The results revealed that the most potentially stable phases predominantly adopted corner-sharing topologies and exhibited relatively low Li-ion diffusivities. In contrast, several metastable edge-sharing structures demonstrated significantly enhanced ionic conductivity. These findings underscore the critical influence of polyhedral connectivity on Li-ion transport and highlight the

effectiveness of our MLIP-assisted CSP approach for discovering high-performance metastable phases within SSEs. Moreover, by analyzing the AIMD trajectories, we were able to rationalize the origin of its superior conductivity from the perspective of Li-ion accessible volume, quantified by packing efficiency—defined as the fraction of the unit cell volume occupied by nonconductive volume—and by the dynamic distortion of the Li-S_4 sublattice, which represents the local environment encountered by migrating Li-ions. These insights emphasize how the crystal structure directly governs transport behavior. Collectively, this study demonstrated the predictive accuracy of our phase-diagram-guided CSP approach, and the mechanistic insight enabled by machine-learning-driven dynamics simulations, elucidating the relationship between crystal structure and ion transport and offering a new perspective for designing high-performance SSEs to advance ASSB technology.

2. COMPUTATIONAL DETAILS

2.1. DFT and MLIP Simulation Parameters. All DFT calculations and molecular dynamics simulations were performed with the Vienna ab initio simulation package (VASP 5.4.4)^{37–40} and a large-scale atomic/molecular massively parallel simulator (LAMMPS)⁴¹ package, respectively. Structural optimization was conducted using the projector augmented wave (PAW)^{42,43} method and the Perdew–Burke–Ernzerhof (PBE) generalized gradient approximation (GGA)⁴⁴ functional. We used a Monkhost-Pack scheme using a k -point mesh with an interval of $0.05 \text{ \AA}^{-1}$ in each primitive lattice vector of the reciprocal space and a plane-wave cutoff of 500 eV.⁴⁵ Lattice constants and internal atomic positions were fully relaxed until the residual forces reached $<0.04 \text{ eV/\AA}$. Spin-polarized calculations were performed for systems containing 3d transition metals. Based on our previous work, we used the same computational guidelines for AIMD simulations in this study, including cell size, spin, time step, and simulation temperature in the NVT ensemble with Nosé–Hoover thermostat.^{46–49} An evolutionary algorithm implemented in the Universal Structure Predictor: Evolutionary Xtallography (USPEX)³³ package was used. The initial structures were generated using random symmetric⁵⁰ and topological⁵¹ structures with a population size of 100 per generation. During this process, new offspring structures are created via heredity, permutation, soft mutation, and lattice mutation, including random and topology-based generators with the antiseed option enabled.⁵² In recent years, considerable efforts have been devoted to applying machine learning-based approaches, to accelerate the discovery of SSE.^{53–55} Specifically, MLIPs have attracted significant attention because they can achieve DFT-level accuracy while significantly reducing computational cost. Among the various types of MLIPs, the moment tensor potential (MTP) was selected owing to its well-balanced trade-off between performance and computational cost.^{56,57} Notably, MTP has been successfully applied to various systems, such as SSEs,^{58–60} alloy,⁶¹ interfacial models,^{62,63} grain boundaries,^{64–66} and CSP studies,^{67,68} demonstrating its versatility and effectiveness. Additionally, MTP has been used as a structural optimizer in conjunction with LAMMPS.^{56,69} The hyperparameters used were w_e : w_f : $w_s = 100$:1:0.1, $R_{\text{cut}} = 5 \text{ \AA}$, and $\text{lev}_{\text{max}} = 20$, which were found to be sufficiently high to significantly impact the accuracy based on our previous research.⁶⁴ Further details of

the MTP are provided in [Supporting Information](#). The overall process for CSP was supported by PyMatgen.⁷⁰

2.2. Generation of Initial Structures for the Training Data Set.

Figure 1 illustrates the workflow of CSP with MTP,

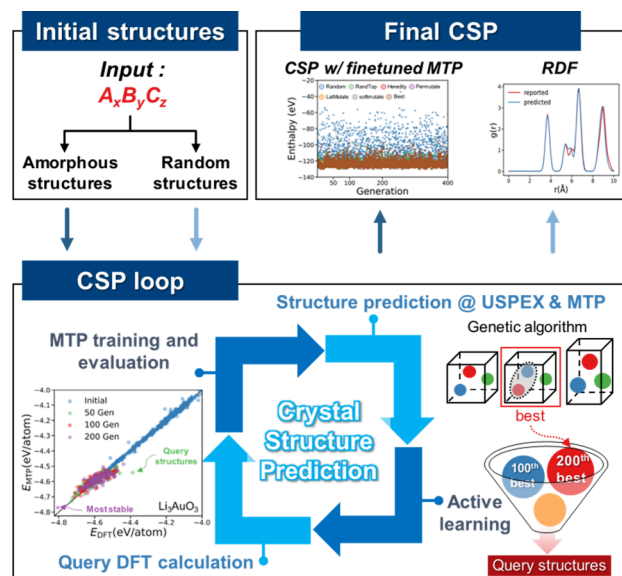


Figure 1. Workflow of CSP with MTP and active learning. The process begins with the generation of initial structures, including amorphous and random structures, corresponding to a target composition $A_xB_yC_z$. The CSP loop iteratively updates the MTP model by fitting to newly selected query structures derived from active learning, ensuring accuracy between DFT and MTP predictions. The predicted structures are validated through energy comparisons and radial distribution function (RDF) analysis against reference structures to assess prediction accuracy.

which consists of three main stages: generation of initial training sets, iterative CSP loop, and final structure prediction. For the accurate prediction of stable phases using MTP, high-quality and structurally diverse training data sets are crucial to ensure DFT-level accuracy in structural optimization for a given composition. However, since the objective is to discover novel crystal structures, conventional frameworks from computational or experimental databases cannot be used directly as training data sets, resulting in inherent data scarcity. To address this challenge, we adopted two complementary strategies to construct the initial training data sets: amorphous and randomly generated structures based on the target compositions. The amorphous structures were generated through a melt-quench-annealing process using AIMD simulations with an NVT thermostat, in accordance with previous CSP studies,^{71–73} to sample the general atomic environments and obtain training data sets on the atomic energies, forces, and stresses. Initially, atoms were placed in a simulation box corresponding to the target composition. The system was heated to 4500 K and equilibrated for 5 ps, which is well above the melting temperature, to ensure the amorphous structure. Subsequently, we assumed 2500 K as an empirically determined melting temperature, which is typically sufficient for most systems to melt, and performed AIMD simulations at this temperature for 10 ps. The resulting molten structure was then cooled down to 300 K at a rate of 200 K/ps through a quenching process. Finally, we carried out an annealing step at 500 K for 4 ps to complete the melt-

quench-annealing cycle and obtain the training set structures. Although these structures provide a broad coverage of the potential energy surface (PES) in an arbitrary coordinate space, they may still leave untrained regions that are insufficient for accurate structural relaxation using MTP. To improve the PES coverage and quality of MTP, randomly initialized structures were also included. Specifically, an initial population of 400 structures was used in USPEX, with atoms arranged in random symmetric or topological configurations. These structures were calculated using single-point DFT calculations to obtain the precise energies, forces, and stresses required for training. By integrating these two approaches, an initial training data set that supported reliable and accurate structural optimization in subsequent CSP processes was constructed.

2.3. Iterative Loop for Crystal Structure Prediction.

Based on the combined training data set, we conducted an iterative CSP loop comprising four major steps: MTP training and evaluation, structure prediction, active learning, and query DFT calculation. Initially, the MTP was trained using specified hyperparameters, and training data set was constructed, enabling efficient structural relaxation that would have required substantial computational time with DFT. Following the initial training, the accuracy of the MTP was evaluated by comparing its predicted energies with the DFT-calculated energies. Structure generation was performed using the USPEX genetic algorithm for the target compositions. In the early iterations of the CSP loop, structures were generated over 50 evolutionary USPEX generations, and MTP was used to relax the candidate structures at each step, allowing the efficient identification of low-energy configurations through an iterative process. However, because the initial MTP training data set consists only of amorphous and randomly generated structures, the accurate optimization of the global minimum structures remains challenging. Therefore, incorporating the new structures discovered within each USPEX generation is essential to further improve the model's accuracy. To this end, high-quality DFT labels (energies, forces, and stresses) for the selected structures are required before updating the training data set. Although MTP drastically reduces the computational burden, acquiring DFT reference data for a large number of predicted structures remains a major computational bottleneck.

To mitigate this issue, we adopted an active learning approach, a machine learning technique that effectively identifies the most informative samples based on uncertainty metrics and selectively adds them to the training data set. This approach significantly enhances model performance with fewer data points and is widely used in MTP-based studies for complex systems such as grain boundaries, interfaces, and CSP. By applying active learning, only high-uncertainty structures (i.e., query structures) are selected from the generated data set. This selective sampling substantially reduces the data set size, requiring DFT evaluation. The query structures are fully relaxed, and single point calculated using DFT. The resulting energies, forces, and stresses are added to the training data set to retrain and refine the MTP. This iterative retraining and evaluation led to progressive improvements in prediction accuracy, as confirmed by the MTP-DFT consistency checks. As the CSP loop progressed, the number of evolutionary generations progressively increased from 100 to 200 and finally to 400, further improving the predictive capability of MTP and facilitating the discovery of stable structures. The search was

terminated when an experimentally reported structure was predicted within the CSP loop. To verify the structural fidelity, symmetry (space group) analysis and radial distribution function (RDF) comparisons against experimental references were conducted. For compositions without experimentally reported structures, the CSP loop was terminated after 400 USPEX generations.

2.4. Final Step of CSP with Fine-Tuned MTP. With the fine-tuned MTP integrated into the CSP workflow, the final round of structure prediction was performed for over 400 evolutionary generations of the targeted compositions. To assess the structural similarities between the newly predicted structures, their space groups and RDFs were compared with their experimentally reported counterparts. In cases where no experimental references were available, DFT-based stability calculations were conducted to rank the predicted structures. This MTP-driven, fully automated CSP framework significantly reduces the computational cost associated with traditional DFT-based CSP and improves the efficiency and scalability of crystal structure discovery across a broad compositional space.

3. RESULTS AND DISCUSSION

3.1. Evaluating Predictive Reliability of MTP-Based CSP. Although previous studies have combined USPEX with MTP for CSP, their efforts primarily focused on unary or binary compositions^{68,74} or have been limited to high-pressure conditions.^{75,76} To evaluate the performance of the CSP method more rigorously, it is essential to predict crystal structures across a broader range of elements and material systems. Particularly, SSEs typically contain mobile ions such as lithium and feature structural frameworks that combine ionic and covalent bonds. This complexity results in a highly intricate PES, which requires thorough validation of MLP-based CSP methods. To this end, we applied CSP to 14 experimentally determined lithium- or sodium-containing structures comprising insulators, conductors, electrodes, and SSEs. A comprehensive summary of the reference structures and CSP results is presented in Figure S1 and Table S1. Figure 2a shows the energy difference (ΔE), calculated using the PBE functional, between the experimental structures and most stable configurations predicted through our CSP method. A negative ΔE indicates that the predicted structure is more stable than the reference. Among the 14 compositions, the proposed method reproduced the experimentally reported structures and predicted a more stable alternative for 12 cases. Furthermore, five candidate compositions exhibited negative ΔE values, indicating that the predicted structures were more favorable than their experimental counterparts.

Despite its outstanding performance, the CSP method failed to reproduce the correct structure or a more stable alternative in the case of LiBiO_3 and $\text{LiGa}(\text{SeO}_3)_2$. To understand the cause of prediction failures, the mean absolute error (MAE) of MTP training was analyzed by comparing the total MAE (MAE_a) and filtered low-energy MAE (MAE_f). The MAE_a for LiBiO_3 is comparable to successful cases, suggesting that more evolutionary steps may be needed. In contrast, the large MAE_f in $\text{LiGa}(\text{SeO}_3)_2$ (80 atoms/unit cell) indicates limited accuracy in low-energy regions due to structural complexity. Additionally, in five cases where CSP discovered structures with lower energy compared with known references, four systems (Li_3PS_4 , Na_3YBr_6 , Na_3PS_4 , and Na_3YCl_6) not only exhibited improved stability but also reproduced experimental structures within

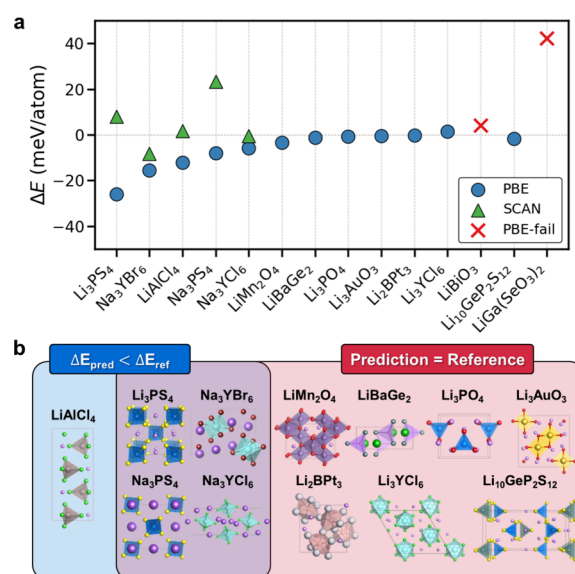


Figure 2. (a) Energy difference (ΔE) between most stable predicted and experimental structures calculated using PBE and SCAN functionals. Blue circles represent results obtained from PBE functional, and green triangles represent those obtained from SCAN. Red crosses indicate cases where PBE failed to identify the correct ground-state structure. (b) Classification of 12 experimentally known structures based on prediction outcomes. Blue and red regions indicate cases where the predicted structure is more stable than the experimental reference and cases where the predicted and experimental structures match, respectively. The overlapping region represents structures for which both the predicted and reference phases were successfully identified.

PBE, as shown in Figure 2b. Their potential validity was confirmed through SCAN calculations, ruling out PBE-related artifacts.^{72,77–79} A detailed analysis of both energetically improved predictions and unsuccessful cases is provided in Supporting Information. Notably, the proposed CSP successfully predicted the $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ structure, which is a quaternary system with 50 atoms per unit cell. Since the CSP complexity increases exponentially with the number of atomic species and total atoms, this result underscores the robustness and scalability of our method. Moreover, the results demonstrate the strong predictive capability of the proposed CSP framework, which not only accurately reproduces known structures but also identifies potentially superior alternatives, highlighting the framework's reliability and efficiency in predicting SSE crystal structures.

3.2. Phase-Diagram-Guided CSP for Li_2SiS_3 -Based Systems. The CSP approach has proven to be an effective tool for exploring novel crystal structures in SSEs, offering a pathway for the discovery of high-performance materials. Because the crystal structure fundamentally determines critical properties such as Li-ion transport, predicting new structures can enhance ionic conductivity beyond conventional doping strategies. Given that CSP generates a vast number of candidate structures, it is not limited to predicting only the most stable configuration but can also identify numerous metastable phases, some of which exhibit high ionic conductivity. This capability expands the search space for promising SSE candidates and enhances their potential for the discovery of novel materials.

To systematically predict promising SSE structures, Li_2SiS_3 was selected as the primary composition for CSP. Li_2SiS_3 was

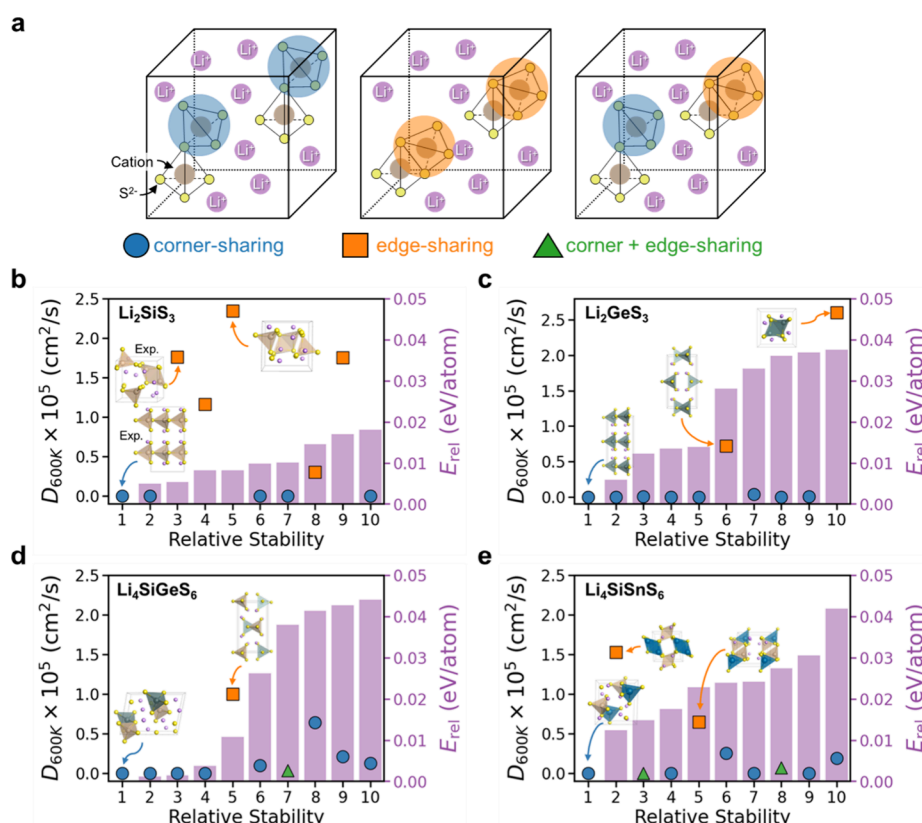


Figure 3. (a) Classification of predicted crystal structures based on polyhedral connectivity: corner-sharing (left, blue), edge-sharing (middle, orange), and mixed corner/edge-sharing (right, green). Li-ion diffusivity ($D_{600\text{ K}}$), obtained from AIMD simulations over 60 ps at 600 K, and relative potential energy (E_{rel}) of predicted 10 most stable structures for (b) Li_2SiS_3 , (c) Li_2GeS_3 , (d) $\text{Li}_3\text{SiGeS}_6$, and (e) $\text{Li}_4\text{SiSnS}_6$. The shape and color of scatter points indicate the polyhedral connectivity types: corner-sharing (blue circles), edge-sharing (orange circles), and mixed corner/edge-sharing (green triangles). The purple bars represent E_{rel} with respect to the most stable predicted structures. The x -axis represents the relative stability ranking, ordered from most stable to least stable. The insets depict representative crystal structures with their corresponding polyhedral connectivity.

used exclusively as a validation benchmark to demonstrate that our MLIP-assisted CSP framework can correctly reproduce both the stable phase and the experimentally known metastable edge-sharing phase. In addition to Li_2SiS_3 , a simple substitution strategy was applied by replacing Si with Ge, which is a Group 14 element commonly used in SSE systems.^{80,81} As Group 14 elements, Ge^{4+} (67 pm) and Si^{4+} (54 pm) exhibit similar chemical properties. However, the larger ionic radius of Ge^{4+} is expected to enhance ion migration by expanding diffusion pathways. In contrast, Sn was excluded from the CSP due to the prior investigation of Li_2SnS_3 as an SSE.⁸² Additionally, CSP has been extended to quaternary systems, which require the identification of optimal structural compositions suitable to experimental synthesis. To achieve this, we adopted a phase-diagram approach to determine the optimal quaternary compositions, including those of Li and S. Previous studies have demonstrated that this approach is effective for selecting favorable compositions from a large number of possible combinations.^{83–86} Phase-diagrams were constructed using the Li, Si, S, and [Ge, Sn] compositions to determine suitable target structures. Next, $\text{Li}_4\text{SiGeS}_6$ and $\text{Li}_4\text{SiSnS}_6$ were selected based on their 1:1 stoichiometric ratio, which lies along the line connecting the SiS_2 and $\text{Li}_4[\text{Ge, Sn}]\text{S}_4$ phases, as illustrated in Figure S2. These well-balanced compositions, derived from the partial substitution of Si in Li_2SiS_3 , were considered theoretically and experimentally feasible for synthesis. Based on these selected four

compositions, CSP was conducted to predict their stable crystal structures. To ensure accuracy, the consistency between MTP and DFT calculations was evaluated using the correlation analysis, as shown in Figure S3. To systematically classify the predicted structures, polyhedral connectivity was used as a criterion, dividing them into three categories: corner-sharing, edge-sharing, and mixed corner/edge-sharing topologies, as shown in Figure 3a. These topological features are directly linked to key material properties such as relative potential energy and ionic conductivity.

Figures 3b–e shows the relative stabilities, polyhedral connectivity types, and Li-ion diffusivities of the predicted structures. Details of the computational procedure for diffusivity are described in Supporting Information, and the mean square displacement (MSD) results for the 10 most stable structures in each composition are shown in Figures S4–S7. Among the three connectivity types, the corner-sharing topologies (blue circles) consistently exhibited the highest stability across all four compositions, likely due to their increased structural flexibility compared with their edge-sharing counterparts (orange squares). In contrast, despite their lower stability, edge-sharing topologies exhibited significantly enhanced Li-ion diffusivity.

To enable a thermodynamic comparison of the phases at 300 K beyond the 0 K DFT calculations, Gibbs free energy calculations were performed. As shown in Figure S8, the Gibbs free energies at 300 K confirm that the representative corner-

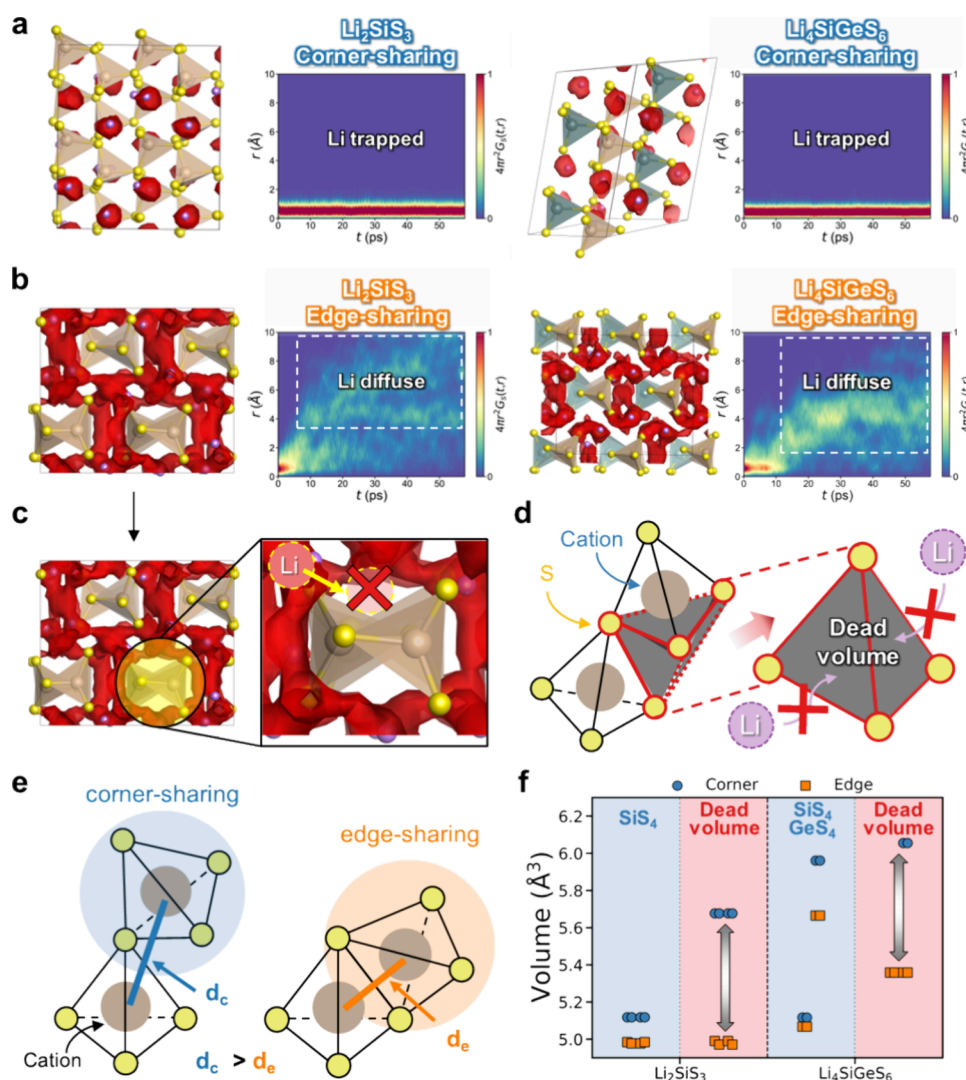


Figure 4. Isosurface of Li-ion probability density distribution and self-part of van Hove correlation function for different polyhedral topologies in Li_2SiS_3 and $\text{Li}_4\text{SiGeS}_6$: (a) corner-sharing and (b) edge-sharing topologies. (c) Enlarged view of Li-ion pathways in edge-sharing topologies. (d) Schematic illustration of dead volume formation in corner-sharing topologies. (e) Comparison of center-to-center distances between adjacent polyhedra in corner- (d_c) and edge-sharing (d_e) topologies. (f) Distribution of polyhedron volume in corner- and edge-sharing topologies. The blue and red shaded regions represent the polyhedron volume and dead volume, respectively.

sharing structures remain thermodynamically more stable than the edge-sharing structures for all compositions, indicating that the stability trend obtained from 0 K DFT potential energies remains valid at room temperature, as detailed in the Supporting Information. To further assess experimental synthesizability, we computed the energy above the convex hull (E_{hull}) for ten representative structures across four compositions. E_{hull} is widely regarded as a reliable indicator of synthetic accessibility in solid-state materials.^{22,87} As summarized in Table S2, all candidate structures exhibit E_{hull} values below 50 meV/atom, indicating a high likelihood of experimental realization. Taken together, these free-energy and E_{hull} results demonstrate that the predicted structures are thermodynamically reasonable at near-ambient and elevated temperatures and are highly likely to be experimentally synthesizable.

Specifically, the CSP framework accurately reproduced the experimentally reported corner-sharing structure as the most stable configuration in Li_2SiS_3 , as shown in Figure 3b. Furthermore, the experimentally observed metastable edge-

sharing structure was successfully recovered and ranked as the third most stable structure, with RDF analysis confirming its structural match (Figure S9). Notably, the Li-ion diffusivity of this edge-sharing structure is at least 2 orders of magnitude higher than that of the corner-sharing structure, aligning well with experimental observations reporting an enhancement of over 3 orders of magnitude in ionic conductivity.³⁶ Beyond the third-ranked structure, other metastable edge-sharing structures also exhibited exceptional Li-ion diffusivity, highlighting the superior ionic transport properties of metastable phases. Importantly, the relative potential energy difference between the most stable corner-sharing and metastable edge-sharing topologies is <0.2 eV/atom, indicating that edge-sharing topologies are sufficiently viable for synthesis. These results demonstrate the effectiveness of CSP in reproducing known structures and uncovering metastable topologies with high ionic conductivities, highlighting the edge-sharing topology as a promising candidate for SSEs. Figure 3c exhibits a similar trend for Li_2GeS_3 , where edge-sharing topologies exhibit superior ionic transport compared to corner-sharing top-

ologies. However, despite their promising ionic mobility, the edge-sharing topologies of Li_2GeS_3 exhibited relatively high formation energies ($E_{\text{rel}} \geq 0.3$ eV/atom), which are significantly higher than those of Li_2SiS_3 , thereby limiting experimental viability. Consequently, all reported Li_2GeS_3 structures to date have been limited to corner-sharing topologies, which inherently offer low ionic conductivity, restricting their viability as SSEs. Nevertheless, the successful implementation of metastable edge-sharing topologies could open new pathways for SSE development by significantly enhancing ionic transport performance.

Figure 3d,e highlight the exceptional Li-ion transport properties of the metastable edge-sharing topologies in $\text{Li}_4\text{SiGeS}_6$ and $\text{Li}_4\text{SiSnS}_6$, underscoring their potential as SSEs. In $\text{Li}_4\text{SiGeS}_6$, the metastable edge-sharing topology (orange square) exhibits superior Li-ion diffusivity compared with other polyhedral topologies, as shown in Figure 3d. Furthermore, its relatively low E_{rel} (≤ 0.1 eV/atom) suggests favorable experimental accessibility, making it a highly promising metastable SSE candidate with excellent ionic transport. In addition, this polyhedral topology is formed through heteroelemental (Si–Ge) linkages rather than homoelemental (Si–Si) connections, which enhances structural disorder and flexibility, thereby improving overall stability. This composition also includes structures featuring mixed corner- and edge-sharing motifs (green triangles). However, their Li-ion diffusivities are markedly lower, highlighting their limited potential for use in SSE applications. A similar trend was observed in $\text{Li}_4\text{SiSnS}_6$, where edge-sharing topologies (orange square) also exhibited high Li-ion diffusivity while maintaining relatively low E_{rel} (≤ 0.3 eV/atom), suggesting higher experimental feasibility compared with Li_2GeS_3 . Notably, $\text{Li}_4\text{SiSnS}_6$ features more frequent occurrences of mixed corner- and edge-sharing motifs (green triangles) compared with other systems. This trend is attributed to the relatively large ionic radius of Sn^{4+} (83 pm), which facilitates more flexible bonding geometries compared with those of Si^{4+} (54 pm) and Ge^{4+} (67 pm). However, mixed topologies consistently showed poor Li-ion transport, similar to what was observed in $\text{Li}_4\text{SiGeS}_6$, as corner-sharing topologies act as bottlenecks, impeding Li-ion migration. Taken together, these findings emphasize that, among the various polyhedral topologies, only the edge-sharing metastable phases can consistently deliver both high ionic conductivity and experimental viability, paving the way for the discovery and development of novel high-performance SSE materials.

3.3. Insights into the Superior Li-ion Diffusivity of Metastable Edge-Sharing Phases. So far, we performed CSP on four sulfide electrolyte compositions, revealing that the edge-sharing metastable phases exhibit favorable Li-ion mobility and can be synthesized more readily when their relative stability falls within a moderate range. Generally, metastable phases exhibit lower energy barriers for Li-ion migration compared with potentially stable phases. However, although metastable phases with other topologies also exist, these structures exhibit significantly lower ionic diffusivity, and the specific reasons for the superior performance of edge-sharing metastable phases remain unclear. Furthermore, a recent study using high-throughput screening of inorganic crystal structure databases revealed that corner-sharing oxide structures achieved moderate ionic conductivities below 1 mS/cm in both theoretical and experimental evaluations.⁸⁸ This

observation contradicts the results obtained by this study, and further investigation is essential to elucidate the mechanisms underlying Li-ion mobility.

To address this, the Li-ion migration behavior in the most stable corner-sharing phase and promising metastable edge-sharing phases of the Li_2SiS_3 and $\text{Li}_4\text{SiGeS}_6$ were analyzed. Figure 4a,b show the isosurfaces (red) of the Li-ion probability densities and the self-part of the van Hove correlation obtained from 60 ps AIMD simulations at 600 K. The probability density results clearly demonstrate that Li ions in the corner-sharing phase remain localized, whereas Li ions in the edge-sharing phase transport readily, exhibiting substantially enhanced migration regardless of composition. Moreover, the van Hove correlation function distinctly highlights the structural differences between the corner- and edge-sharing phases during Li-ion diffusion. In the corner-sharing phase, only a strong peak below 2 Å was observed, indicating that Li ions remained confined during the 60 ps simulation. In contrast, the edge-sharing phase initially exhibited a peak below 2 Å during the first 10 ps, followed by a well-distributed pattern across longer distances, suggesting that Li-ions were not trapped and migrated efficiently.

From the Li-ion transport analysis, no significant Li-ion migration was observed in the interstitial spaces between connected polyhedra in either the corner- or edge-sharing topologies as shown in Figure 4c. These regions inevitably form pseudopolyhedron voids due to the repulsive forces between the cations at the polyhedron centers (Si^{4+} , Ge^{4+} , and Sn^{4+}) and the Li ions, effectively hindering Li-ion diffusion. These areas, referred to as “dead volume” appear as empty regions within the crystal structure and are inaccessible for Li-ion migration, as illustrated in Figure 4d. Since these volumes limit the effective migration space for Li-ions and influence the overall ionic conductivity, a proper evaluation of ion transport should consider both the dead volume and total lattice volume. However, most previous studies have overlooked these regions. To gain a more comprehensive understanding of the Li-ion conductivity in SSEs, it is crucial to account for the impact of this dead volume on the ionic transport properties.

Figure 4e demonstrates that the edge-sharing topology features a shorter distance between the central cations of adjacent polyhedra compared with the corner-sharing topology, resulting in a substantial reduction in dead volume. Furthermore, due to the shared edges, each anion is influenced by the electrostatic forces from two neighboring cations, which further compress the polyhedron and reduce its volume. To verify this, we compared the polyhedron volume and dead volume in four crystal structures previously analyzed for the Li-ion probability density, as shown in Figures 4f and S10. The results clearly show that, in both compositions, the edge-sharing topology exhibited a smaller polyhedron volume and dead volume, with a more significant reduction observed in the dead volume. Therefore, the superior Li-ion mobility of the edge-sharing phase can be attributed to a reduction in the polyhedron volume, while the notable decrease in dead volume contributes to an expanded Li-ion migration space within the crystal structure.

Based on the dead volume concept, we aimed to systematically analyze the key structural factors influencing ion diffusion in SSEs by quantitatively examining the exceptional Li-ion diffusivity observed in the predicted metastable edge-sharing phases. To enable efficient Li-ion transport within a crystal structure, the presence of sufficient

free volume is essential. However, rather than simply having large spaces, it is crucial that these spaces are effectively accessible for Li-ion migration to ensure high ionic conductivity. Furthermore, to readily assess the performance of various structures, it is important to quantify the effective volume available for Li-ion transport. To achieve this, we introduce the packing ratio (α), which quantifies proportion of the crystal volume occupied by structural features that hinder Li-ion mobility, thereby establishing a relationship between the lattice volume, dead volume, and ionic conductivity, as detailed in Supporting Information. This parameter represents the fraction of lattice space available for Li-ion transport, with a lower α value indicating a larger effective migration space within the same volume and enhanced ionic conductivity. Although the packing ratio offers valuable insights into the available migration space, it cannot fully describe the local environment encountered by the migrating Li-ions. Therefore, a complementary analysis from the perspective of Li-ion dynamics is essential for a more comprehensive understanding of ion transport in metastable edge-sharing phases, given the critical role of Li-ion mobility in determining ionic conductivity.

To this end, the investigation focused on the Li–S₄ sublattice, which is a fundamental structural moiety that directly interacts with Li-ions. Figure S11a illustrates the Li–S₄ sublattice, wherein sulfur anions form a tetrahedral environment around Li ions. These anions form attractive interactions with Li ions, and subsequently influence ionic conductivity. Specifically, the larger distances between the Li ions and polyhedral vertices (S^{2−}) suggest that a large Li–S₄ sublattice volume within the lattice enhances Li-ion mobility by reducing electrostatic interactions. Additionally, previous studies have indicated that significant polyhedral distortion reduces the activation energy for Li-ion diffusion by creating a smoother Li-anion interaction energy landscape, thereby lowering the Li-ion migration energy barrier, as shown in Figure S11b.^{88,89} To validate this relationship, AIMD simulations were performed to analyze the Li–S₄ sublattice volume in the ten most stable predicted structures, as discussed in the previous section. Furthermore, to quantitatively assess these distortions, continuous symmetry measure (CSM) analysis was conducted based on AIMD simulations, where higher CSM values indicated greater deviations from the ideal symmetry in the Li–S₄ sublattice. Further information on CSM is provided in Supporting Information.

Figure 5 presents the packing ratio (α), Li–S₄ sublattice volume, and CSM for the 10 most stable predicted structures of the Li₂SiS₃ and Li₄SiGeS₆ compositions. Specifically, Figure 5a shows that the edge-sharing phases in the Li₂SiS₃ composition (green bars) exhibit significantly lower α values, possibly owing to the reduced dead volume in these structures. This trend confirms that the α value effectively captures the structural characteristics of the edge-sharing topology with high packing efficiency, highlighting the importance of volumetric factors in Li-ion mobility. Additionally, both the Li–S₄ sublattice volume and the CSM values of the edge-sharing phases are higher than those of the corner-sharing phases, indicating that the Li–S₄ sublattice in the edge-sharing phases are larger and more distorted (colored circles). This provides a clear structural rationale for the superior ionic conductivity observed in edge-sharing phases. In contrast, corner-sharing phases exhibit relatively higher α values, with the Li–S₄ sublattice volume and CSM confined to a narrow

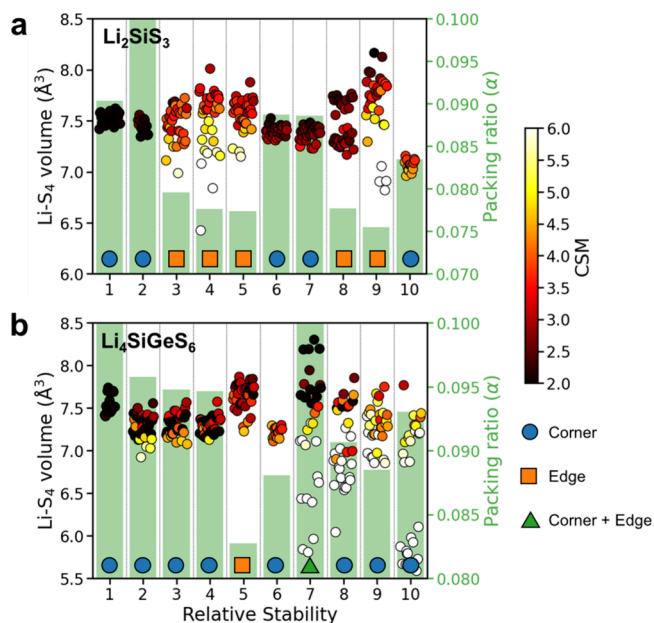


Figure 5. Analysis of 10 most stable predicted structures for (a) Li₂SiS₃ and (b) Li₄SiGeS₆ compositions in terms of the packing ratio (green bars), average Li–S₄ polyhedron volume (height of circles), and CSM (circle color). Data are extracted from AIMD simulations at 600 K over 60 ps. The packing ratio (α) quantifies the nonconductive volume fraction, while CSM reflects the degree of distortion in the Li–S₄ polyhedron. The shape of the base markers represents polyhedral connectivity: blue circles (corner-sharing), orange squares (edge-sharing), and green triangles (corner/edge-sharing).

range, suggesting limited structural flexibility and restricted Li-ion mobility. Notably, the most stable corner-sharing phases exhibited exceptionally low CSM, reflecting pronounced structural rigidity and negligible Li-ion displacement, which is consistent with the Li-ion probability density shown in Figure 4a and the experimentally observed extremely low ionic conductivity.

Contrasting trends between the corner- and edge-sharing phases appear consistently in Figures 5b and S12. Specifically, the edge-sharing phases in both compositions exhibit significantly lower α compared with other phases, while their Li–S₄ sublattice volume and CSM values are relatively higher. These features correlate well with the high Li-ion diffusivity observed in the AIMD simulations, further validating the proposed structural descriptors as key indicators for predicting ionic conductivity and reinforcing the superior transport properties of edge-sharing phases. In conclusion, the combined analysis of the packing ratio, Li–S₄ sublattice volume, and CSM revealed that edge-sharing structures provide a larger Li-ion migration space by reducing dead volume and weakening the electrostatic interactions between Li-ions and anions (S^{2−}). These structural changes result in an expanded Li–S₄ sublattice volume and higher distortion, both of which contribute to enhanced Li-ion conductivity. This study offers valuable insights into the design of SSEs, deepens the theoretical understanding of edge-sharing phases, and establishes reliable descriptors for evaluating the Li-ion mobility in novel candidate materials. Moreover, these findings lay a solid foundation for the theoretical design of advanced next-generation SSEs with unique crystal structures, thereby accelerating the discovery and development of advanced materials for high-performance all-solid-state batteries.

4. CONCLUSIONS

This study aimed to predict novel SSE structural phases to overcome the limitations of existing crystal structures through a theoretical approach. To this end, CSP process integrated with the machine-learning MTP was developed and validated. The proposed CSP process reproduces the experimentally reported crystal structures of 12 different compositions. Building on this predictive accuracy, phase-diagram-guided CSP approaches identified Li_2SiS_3 , Li_2GeS_3 , $\text{Li}_4\text{SiGeS}_6$, and $\text{Li}_4\text{SiSnS}_6$ compositions as plausible SSE candidates, leading to the CSP of their novel crystal structures. The predicted structures obtained from CSP were subsequently subjected to AIMD simulations, which revealed that, while the corner-sharing phases exhibited the highest potential stability with extremely low Li-ion diffusivity, the metastable edge-sharing phases demonstrated significantly enhanced Li-ion transport. The Li-ion probability density analysis confirmed this trend, indicating that Li-ion transport was suppressed in the interstitial space between the connected polyhedra. This void space, referred to as “dead volume” hindered Li-ion diffusion due to repulsive forces between central cations (Si^{4+} , Ge^{4+} , and Sn^{4+}) and Li-ions. Structural investigations further revealed that the edge-sharing topologies significantly reduced the dead volume, thereby providing a substantially larger effective space for Li-ion diffusion. To systematically and quantitatively evaluate the key factors governing Li-ion migration, the packing ratio (α), Li– S_4 sublattice volume, and CSM of the Li– S_4 sublattice were analyzed. The results demonstrated that the edge-sharing phases exhibit a significantly lower packing ratio, larger Li– S_4 sublattice volume, and greater distortion of the Li– S_4 sublattice, all of which are positively correlated with enhanced ionic transport. Overall, this study provides valuable insights into the rational design of next-generation SSEs using phase-diagram-guided and MLIP-assisted CSP approach. Particularly, the importance of metastable edge-sharing topology is highlighted, and the key structural descriptors, packing ratio, Li– S_4 sublattice volume, and CSM of the Li– S_4 sublattice are established as predictive indicators of high Li-ion diffusivity in solid-state electrolytes.

■ ASSOCIATED CONTENT

Data Availability Statement

All data supporting the findings of this study are available in the [Electronic Supporting Information](#). Raw data is available at: [10.6084/m9.figshare.29468165.v4](https://doi.org/10.6084/m9.figshare.29468165.v4).

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.5c15665>.

Moment tensor potential, CSP, diffusivity, packing ratio, CSM analysis, and calculated properties (PDF)

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Notes

The authors declare no competing financial interest.

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